A
Fibroblast
Atlas map: 40,000 stratified cells
Myfibro
eFibro
iFibro
qFibro
UMAP2
UMAP1
B
Endothelial
Atlas map: 40,000 stratified cells
ArtEC
CapEC
EndoMT
LEC
MCs
VenEC
UMAP2
UMAP1
C
Subtype hUSI; donor unit
qFibro_CLU_bone (n=25)
Myfibro_CTSA_skin (n=11)
qFibro_MGP_eye (n=1)
eFibro_COL6A1_adipose (n=1)
iFibro_CD74_pan (n=81)
eFibro_DCN_uterus (n=10)
qFibro_CLU_pan (n=159)
eFibro_IGFBP7_pan (n=74)
eFibro_SPARC_pan (n=48)
eFibro_DCN_heart (n=78)
qFibro_APOD_breast (n=75)
qFibro_CFD_pan (n=150)
eFibro_COL1A2_pan (n=321)
Myfibro_TAGLN_pan (n=366)
eFibro_DCN_pan (n=203)
Myfibro_KRT8_pan (n=30)
iFibro_ARHGDIB_pan (n=25)
Myfibro_TPM1_pan (n=71)
eFibro_COL3A1_pan (n=70)
eFibro_HNRNPA1_pan (n=101)
Myfibro_TUBB2B_pan (n=123)
iFibro_CD74_bone marrow (n=45)
eFibro_COL1A1_pan (n=402)
Myfibro_CALD1_pan (n=219)
Myfibro_CDKN2A_brain (n=11)
eFibro_COL3A1_lung (n=30)
Myfibro_MYH11_uterus (n=25)
Myfibro_QKI_eye (n=53)
Myfibro_QKI_brain (n=169)
Myfibro_CALD1_kidney (n=63)
eFibro_COL1A1_lung (n=27)
0.2
0.3
0.4
0.5
0.6
0.7
0.8
hUSI (native scale)
Equal donor mean ± 95% bootstrap CI
D
Subtype hUSI; donor unit
EndoMT_GNG11_vessel (n=3)
VenEC_TAGLN_pancreas (n=40)
EndoMT_VIM_pan (n=49)
VenEC_DCN_breast (n=2)
CapEC_DCN_breast (n=6)
CapEC_TAGLN_pan (n=2)
CapEC_TAGLN_prostate (n=18)
CapEC_TM4SF1_breast (n=12)
EndoMT_CFD_pan (n=80)
EndoMT_TM4SF1_pan (n=33)
VenEC_TAGLN_pan (n=15)
VenEC_CFD_pan (n=41)
ArtEC_CXCL12_pan (n=125)
EndoMT_EFEMP1_skin (n=1)
CapEC_TM4SF1_pan (n=260)
VenEC_FLT1_pan (n=134)
LEC_EFEMP1_pan (n=80)
VenEC_EPCAM_pan (n=9)
VenEC_CLU_pan (n=152)
CapEC_GNG11_pan (n=43)
EndoMT_TGFBI_pan (n=1)
EndoMT_COL1A1_pan (n=6)
VenEC_TM4SF1_breast (n=8)
EndoMT_CFD_bone marrow (n=39)
EndoMT_TPM2_pan (n=46)
EndoMT_PTN_pan (n=87)
CapEC_FLT1_uterus (n=7)
EndoMT_CLU_eye (n=12)
MCs_TPM2_pan (n=3)
EndoMT_CD3D_pan (n=7)
VenEC_CLU_eye (n=101)
ArtEC_FLT1_brain (n=38)
ArtEC_ARL15_pan (n=5)
ArtEC_FLT1_pan (n=90)
ArtEC_THY1_pan (n=48)
EndoMT_TAGLN_pan (n=46)
ArtEC_HSPG2_brain (n=1)
0.2
0.3
0.4
0.5
0.6
0.7
0.8
hUSI (native scale)
Equal donor mean ± 95% bootstrap CI
E
Within-subtype adjusted age effects
Myfibro_KRT8_pan (n=18, q=0.93)
eFibro_HNRNPA1_pan (n=33, q=0.074)
eFibro_COL3A1_lung (n=21, q=0.57)
qFibro_CLU_pan (n=35, q=0.69)
qFibro_CLU_bone (n=22, q=0.074)
eFibro_COL1A2_pan (n=71, q=0.69)
Myfibro_QKI_brain (n=41, q=0.93)
eFibro_COL1A1_pan (n=98, q=0.99)
Myfibro_TAGLN_pan (n=62, q=0.93)
qFibro_CFD_pan (n=29, q=0.96)
eFibro_IGFBP7_pan (n=17, q=0.93)
Myfibro_CALD1_pan (n=44, q=0.69)
eFibro_DCN_heart (n=23, q=0.93)
eFibro_SPARC_pan (n=23, q=0.69)
Myfibro_CALD1_kidney (n=33, q=0.69)
−2
−1
0
1
2
3
4
hUSI slope per year of chronological age
F
Within-subtype adjusted age effects
EndoMT_TPM2_pan (n=25, q=1)
ArtEC_FLT1_brain (n=30, q=1)
VenEC_TAGLN_pancreas (n=36, q=1)
CapEC_TM4SF1_pan (n=47, q=1)
VenEC_CLU_eye (n=38, q=1)
ArtEC_FLT1_pan (n=22, q=1)
−2
−1
0
1
2
3
4
hUSI slope per year of chronological age
G
SID recommendations
Not a senescence score
Myfibro_CALD1_kidney
Myfibro_CALD1_pan
Myfibro_CDKN2A_brain
Myfibro_CTSA_skin
Myfibro_KRT8_pan
Myfibro_MYH11_uterus
Myfibro_QKI_brain
Myfibro_QKI_eye
Myfibro_TAGLN_pan
Myfibro_TPM1_pan
Myfibro_TUBB2B_pan
eFibro_COL1A1_lung
eFibro_COL1A1_pan
eFibro_COL1A2_pan
eFibro_COL3A1_lung
eFibro_COL3A1_pan
eFibro_COL6A1_adipose
eFibro_DCN_heart
eFibro_DCN_pan
eFibro_DCN_uterus
eFibro_HNRNPA1_pan
eFibro_IGFBP7_pan
eFibro_SPARC_pan
iFibro_ARHGDIB_pan
iFibro_CD74_bone marrow
iFibro_CD74_pan
iFibro_CD74_skin
qFibro_APOD_breast
qFibro_CFD_pan
qFibro_CLU_bone
qFibro_CLU_pan
qFibro_MGP_eye
SID1
SID2
SID3
SID4
SID5
SID6
Each SID standardized across subtypes; donor means
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
Column z score
H
SID recommendations
Not a senescence score
ArtEC_ARL15_pan
ArtEC_CXCL12_pan
ArtEC_FLT1_brain
ArtEC_FLT1_pan
ArtEC_HSPG2_brain
ArtEC_THY1_pan
CapEC_DCN_breast
CapEC_FLT1_uterus
CapEC_GNG11_pan
CapEC_TAGLN_pan
CapEC_TAGLN_prostate
CapEC_TM4SF1_breast
CapEC_TM4SF1_pan
EndoMT_CD3D_pan
EndoMT_CFD_bone marrow
EndoMT_CFD_pan
EndoMT_CLU_eye
EndoMT_COL1A1_pan
EndoMT_EFEMP1_skin
EndoMT_GNG11_vessel
EndoMT_PTN_pan
EndoMT_TAGLN_pan
EndoMT_TGFBI_pan
EndoMT_TM4SF1_pan
EndoMT_TPM2_pan
EndoMT_VIM_pan
LEC_EFEMP1_pan
MCs_TPM2_pan
VenEC_CFD_pan
VenEC_CLU_eye
VenEC_CLU_pan
VenEC_DCN_breast
VenEC_EPCAM_pan
VenEC_FLT1_pan
VenEC_TAGLN_pan
VenEC_TAGLN_pancreas
VenEC_TM4SF1_breast
SID1
SID2
SID3
SID4
SID5
SID6
Each SID standardized across subtypes; donor means
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
Column z score
I
Subtype continuous evidence comparison
Exploratory
qFibro_CLU_pan
Myfibro_TAGLN_pan
eFibro_COL1A1_pan
Myfibro_CALD1_pan
eFibro_COL1A2_pan
eFibro_HNRNPA1_pan
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
Mean signed stable EN coefficient
0.00
0.05
0.10
0.15
0.20
0.25
Adjusted hUSI chronological-age slope
J
Subtype continuous evidence comparison
Exploratory
VenEC_CLU_eye
CapEC_TM4SF1_pan
EndoMT_TPM2_pan
0.08
0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
Mean signed stable EN coefficient
0.00
0.05
0.10
0.15
0.20
Adjusted hUSI chronological-age slope
K
Stable-feature enrichment
Exploratory functional candidates
Reactome: Mitochondrial RNA Degradation
eFibro_COL1A2_pan, 5 genes
Reactome: rRNA Processing in the Mitochondrion
eFibro_COL1A2_pan, 5 genes
Reactome: FASTK Family Proteins Regulate Processing and Stability of Mitochondrial RNAs
eFibro_COL1A2_pan, 5 genes
Reactome: rRNA Processing
eFibro_DCN_pan, 6 genes
Reactome: tRNA Processing in the Mitochondrion
eFibro_COL1A2_pan, 5 genes
Reactome: tRNA Processing
eFibro_COL1A2_pan, 5 genes
Hallmark: TNF-alpha Signaling via NF-kB
eFibro_COL1A1_pan, 22 genes
Hallmark: Epithelial Mesenchymal Transition
Myfibro_TAGLN_pan, 19 genes
Reactome: Respiratory Electron Transport
eFibro_COL1A2_pan, 4 genes
Reactome: Scavenging by Class A Receptors
Myfibro_TAGLN_pan, 6 genes
Reactome: Metabolism of RNA
eFibro_DCN_pan, 6 genes
GO_BP: Negative Regulation of Protein Modification by Small Protein Conjugation or Removal (GO:1903321)
eFibro_COL1A1_pan, 7 genes
GO_BP: Negative Regulation of Protein Ubiquitination (GO:0031397)
eFibro_COL1A1_pan, 7 genes
Reactome: Crosslinking of Collagen Fibrils
Myfibro_TAGLN_pan, 5 genes
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
−log10(BH q); full lineage testing family
L
Stable-feature enrichment
Exploratory functional candidates
Reactome: Influenza Viral RNA Transcription and Replication
EndoMT_TPM2_pan, 11 genes
Reactome: Formation of a Pool of Free 40S Subunits
EndoMT_TPM2_pan, 10 genes
KEGG: Ribosome
EndoMT_TPM2_pan, 10 genes
Reactome: GTP Hydrolysis and Joining of the 60S Ribosomal Subunit
EndoMT_TPM2_pan, 10 genes
Reactome: Viral mRNA Translation
EndoMT_TPM2_pan, 10 genes
Reactome: Nonsense Mediated Decay (NMD) Independent of the Exon Junction Complex (EJC)
EndoMT_TPM2_pan, 10 genes
Reactome: L13a-mediated Translational Silencing of Ceruloplasmin Expression
EndoMT_TPM2_pan, 10 genes
GO_BP: Cytoplasmic Translation (GO:0002181)
EndoMT_TPM2_pan, 10 genes
Reactome: Peptide Chain Elongation
EndoMT_TPM2_pan, 10 genes
Reactome: Eukaryotic Translation Initiation
EndoMT_TPM2_pan, 10 genes
Reactome: Eukaryotic Translation Termination
EndoMT_TPM2_pan, 10 genes
Reactome: Cap-dependent Translation Initiation
EndoMT_TPM2_pan, 10 genes
Reactome: Influenza Infection
EndoMT_TPM2_pan, 11 genes
Reactome: rRNA Processing in the Nucleus and Cytosol
EndoMT_TPM2_pan, 10 genes
2.5
5.0
7.5
10.0
12.5
15.0
17.5
−log10(BH q); full lineage testing family